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Epidemiology of Communicable Diseases

**by
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The objectives of this Lecture are to Identify :

- 1-Components of communicable diseases
- 2-The factors that affect infectious diseases in order to success & establish
- 3-Epidemiologic classification that based on the methods of transmission of the diseases



- 4-Host factors that the transmission of the agent to a new host depend on
- 5-The definition of the carrier and its important
- 6-Mode of transmission of the diseases
- 7-Time periods of infection and disease
- 8-Types of epidemics



Epidemiology of communicable diseases

Agent, host, environment

Infectious Disease:- any disease caused by infective agent

Communicable Disease :- it is an infectious disease , occurs when there is a transmission of the disease from one person to another .



Components of communicable diseases :

There are 3 important components in communicable disease

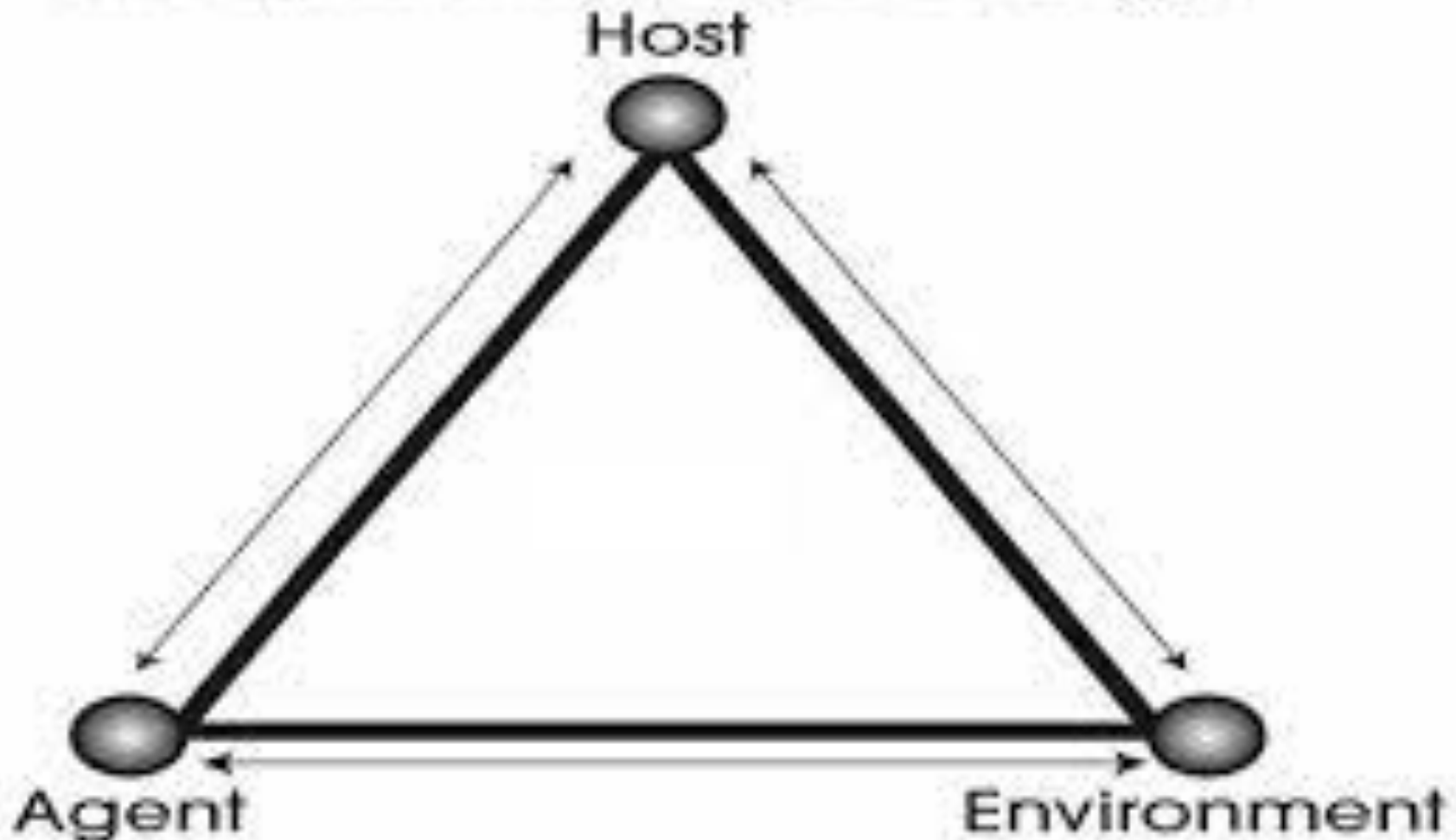
1- agent

2- host

3- environment

They are also called **Epidemiology Triad**

The Epidemiologic Triangle





There is another way to describe a problem of infectious & communicable disease by

Personage, sex , race , etc

Placegeographical distribution

Timewhen the disease occurs

or increases in rate

- in winter diphtheria
- in spring measles & mumps
- in summer cholera & typhoid



**When infection occurs it has 2 forms
either :**

1- Sub clinical

2- Clinical (infectious disease with sign & symptoms)



The infectious disease in order to success to establish there must be the following :-

communicable disease

- 1- The agent should be multiplying**
- 2- The agent should have a susceptible host**
- 3- Should exist from that host**
- 4- Should infect another host**



Epidemiologic classification:
based on **the methods transmission**
and control(fecal-oral diseases,
insect-borne diseases)

air-borne, blood-borne, food-borne
↓
flu, chickenpox, measles
HIV, HBV, HCV
↓
cholera, typhoid, dysentery



Characteristics of communicable diseases

Factors are involved in the causation of communicable diseases:

The agent:

Communicable diseases are caused by an infectious agent or its toxic products.



Agent can be:-

- Viruses
- Bacteria
- Protozoa
- Helminthes → worms
- Fungus



The host

If the agent is transmitted to the new host, its success also depends on a number of host factors:

1. Defense mechanisms: skin, mucous membranes , inflammatory reaction
2. Immunity: cellular and humeral, ^{Role of secondary immune response} acquired from previous exposure or vaccination



→ since Ab needs amino-acid aka protein

3.Nutrition: Malnutrition makes person more susceptible to infection or its complications e.g. measles.

4.Fatigue may weaken immunity and make person more susceptible to infection.

5.Pregnancy:

e.g: pregnant women are more liable to sever malaria

6. Genetics:



Reservoir:-

Is a place where infecting agent can multiply & survive then can infect other host. Some M.O. has more than one reservoir . *↳ Micro-organisms*

The reservoir divided into :

- 1-Human reservoir
- 2-Animal reservoir
- 3-Non-living reservoir



1-Human reservoir

A-Active case → infected people have *sign/symptom*

B-Carrier → they have *micro-organism without symptoms, pathogen in latent phase*

e.g in case of cholera active case with watery diarrhea while the carrier is asymptomatic



Carrier

Human get M.O. but with no signs & symptoms he can transmit the infection to another people (more dangerous)

The carrier is more important because :-

1- Certain diseases have number of carriers more than sick people as meningitis 90% carriers & 10 % with signs & symptoms , in cholera the carriers are more than the symptomatic patients .



2- Carriers have no symptoms & signs

3- Some of the diseases have chronic carriers as Salmonella patient continuous or intermittent spread disease to the community .



Carrier can be divided into :-

1- Temporary Carrier

a-Incubator carrier *→ during disease*

b-Convalescent carrier *→ healing/recovery*

c-Post – convalescent carrier *→ still recovered with some longer pathogen left in patient*

2- Chronic Carrier



2-Animal Reservoir :-

Some agents effect man only , others effect animals can also transmitted to the human .

Zoonotic Disease:-

Is the disease which is transmitted from the vertebrae animals to the human under certain environment as rabies, Anthrax



3- Non Living Reservoir :-

As soil , the M.O. protect itself by several mechanisms → ex: by making spores



Factors That Affect The Host Immunity (Host Factor)

1-Age *in children measles & chickenpox*
in Adult STD

2-Sex

3-Pregnancy

4-Nutrition

5-Trauma and Fatigue

6-Certain infection that suppress immunity
as HIV

7-Systemic Diseases: like diabetes & other
immuno-suppressive diseases.



ENVIRONMENTAL FACTORS :-

- 1-The transmission of the infectious agent to the host takes place within the environment.
- 2-Extent of contact between the agent and host is determined by many environmental factors.
- 3-Conditions that favor more frequent and close contact between the host and the agent favor occurrence of disease.



These are affected by:

- Patterns of mixing *→ overcrowding ↑ risk of air-borne disease*
- Poverty *→ cause malnutrition*
- Education
- Environmental sanitation
- Climate



Mode of transmission :-

Is the mechanism by which M.O. transfer from one person to another or from reservoir to human It can be divided into :

1- Contact

- Directas flue or sexual transmitted diseases
- Indirect...by paper , pen , etc....



2-Inhalation

- Droplets(direct)
- Air born(indirect)

3- Ingested food & water

4- Penetration of the skin

By vector

M.O. directly penetrates the skin

Contaminated wound

5-Transplacental

↳ direct



Conclusions

1-Components of communicable diseases are:

Agent, Host and Environment

2-Infectious diseases in order to success & establish:

The agent should be multiplying, have a susceptible host ,exist from that host and infect another host

3-Epidemiologic classification that based on the methods of transmission of the diseases are:

Contact , Inhalation , Ingested food & water,
Penetration of the skin by vector and transplacental



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Time periods of infection and disease



Time Periods of Infection

The course of an infectious disease can be divided to the following time periods of infection:

1-Time of infection: The point at which the infectious agent enters the susceptible individual



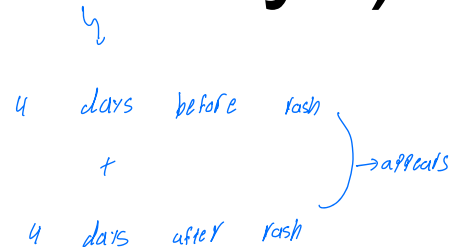
could have a range (chat 6th)

2-Latent period: the period from the time of infection until the infected individual is able to transmit the infection (e.g. for measles it is usually less than the incubation period, 7-18 days).



3-Infectious period or communicable period:

The period during which the infected person is able to transmit the infectious agent to other individuals (measles: around 8 days).





- In diphtheria and streptococcal infections the infectious period starts from day of infection → very short latent period
- in measles and hepatitis it starts before symptoms
- in cholera after symptoms.

↳ by direct basally fecal contaminants contaminate food/water and transmit it



4-Non-infectious period:

when the infectious period ends.

This can occur because the individual recovers and becomes immune; recovers and returns to the susceptible state; or dies from the infection.

Need getting it again

*u
There's an (!!) hole
u
You may get it again*



Time Periods of Disease *→ what's sign and symptoms*

Infection is not equal to disease; a person may have the infection but may not show the symptoms and signs of the disease.

We only say that someone has an infectious **disease** when they have symptoms and/or signs associated with infection.



Some infected people do not develop clinical disease.

We say that these people have **asymptomatic infection**.

so they have time period
of infection no time period
of disease

The proportion of infected people who develop clinical disease varies according to the nature of infective agent.



For some infectious agents, people with asymptomatic infection may still be capable of transmitting the agent to other people.

In these circumstances, we should take asymptomatic cases into account when we are studying the transmission of an infection in a population.



The course of an infectious disease can be divided into the following time periods:

- **Time of infection:** when the person is exposed and gets the infection
- **The incubation period:** the period from the time of infection to the development of symptoms of the disease.



The incubation period varies from one disease to another and for each disease has got a range from the minimum incubation period to the maximum incubation period (e.g. measles: 7-18 days, average 10).



A number of factors affect the incubation period such as :

- The nature of the pathogen *→ virulent pathogen less IP*
- The infective dose *→ amount of pathogen entering the higher it's the lower IP*
- The portal of entry *→ less protected portal of entry less IP*
- The host immune response.

↳ weak immune system less IP



- **Symptomatic period:** the period of time during which the patient has symptoms.

The duration of this period may vary greatly between individuals depending on the severity of the disease.



- **Asymptomatic period:** the symptomatic period ends when the infected person becomes non-symptomatic.

This can occur because the individual recovers from symptoms or dies.

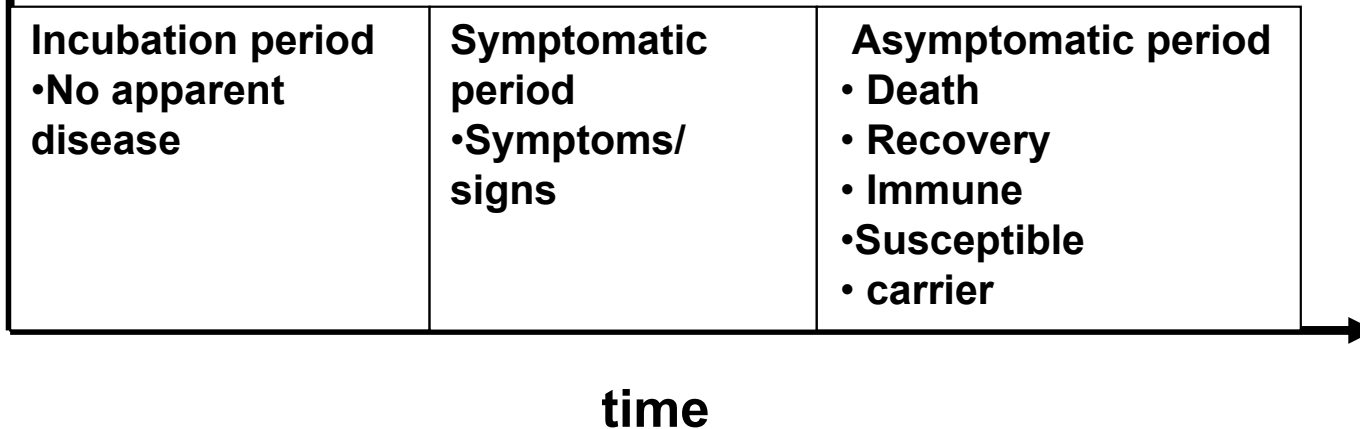


- An individual who recovers from symptoms may be no longer infectious, and can be either **immune** to the infection (if the Infection confers immunity)
or return to the susceptible state (if the infection does not confer immunity).



Time periods of disease

**Time of
infection**





Relation between time periods of disease and infection

The relation between the time periods of infection and the time periods of disease for each infectious agent is extremely important for controlling the transmission of infection. It is especially important to be aware of the length of the latent period and the incubation period of the disease.

→ communicable disease



For example for measles, the latent period is generally shorter than the incubation period. → kinda appearance of rash

This means that most people with measles are infectious *before* they develop symptoms. So, isolating children with symptoms can not prevent all transmission, as some transmission will occur before symptoms have appeared.



Types of epidemics

a sudden unexpected increase in number of cases in a local area basically local outbreak, when it becomes global / international outbreak it's called pandemic

An **epidemic** means occurrence of an infectious disease in a community clearly in excess of what is normally expected or occurrence of a new infectious disease.

When the disease rate is stable in the community is called **endemic**. *constant number of cases every year*



In case of rare disease even appearance of few cases may be considered an epidemic. But when the disease is common such as diarrhea, the number of cases must be substantially increased to be considered an epidemic.



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Types of epidemics:

- 1-Common source epidemic
- 2-Propagated epidemic



Types of epidemics:

single source
1-Common source epidemic: when all infections happen from the same source
e.g. food, water.

Common source epidemic is further divided to:



A-Point source:

when all infections happen from a single exposure.

e.g. when 50 people eat a meat brought in a party and 10 of them get diarrhea next morning.

The epidemic curve will be sharp and bell shaped.



→ exposed multiple times

B-Extended source outbreak:
when there is repeated exposure
to the same source of infection
such as a contaminated water well.
The epidemic curve will be broad.

*↓
presence of plateau line*



2-Propagated epidemic:

an epidemic in which the infectious agent continues to spread from persons to person. In this case the infection will last for a longer time and there will be several peaks in the number of the cases.

kinda multiple source

the curve will gradually increase and gradually decrease

Measles and chickenpox can start as a common source epidemic in a school and continue as a propagated epidemic.

children with symptoms can not prevent all transmission, as some transmission will occur before symptoms have appeared.

Types of epidemics

An **epidemic** means occurrence of an infectious disease in a community clearly in excess of what is normally expected or occurrence of a new infectious disease. When the disease rate is stable in the community is called **endemic**. In case of rare disease even appearance of few cases may be considered an epidemic.

Types of epidemics:

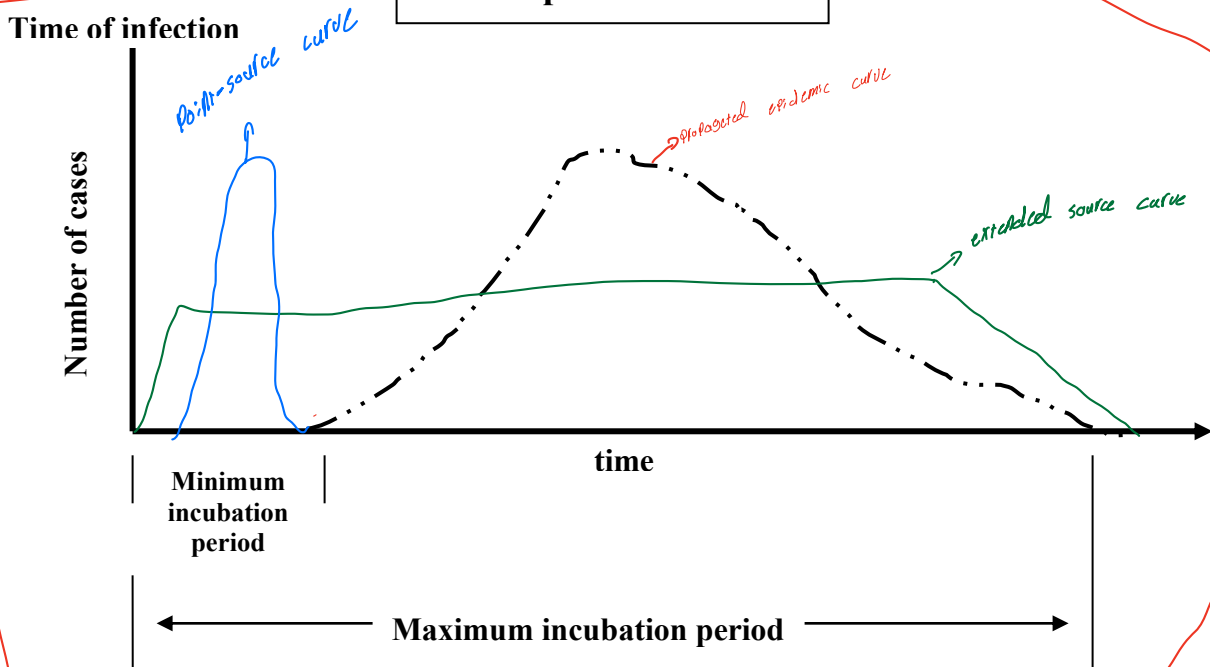
(single source) **1-Common source epidemic:** when all infections happen from the same source e.g. food, water. Common source epidemic is further divided to:

once **a-Point source:** when all infections happen from a single exposure. e.g. when 50 people eat a meat brought in a party and 10 of them get diarrhea next morning. The epidemic curve will be sharp and bell shaped.

multiple times **b-Extended source outbreak:** when there is repeated exposure to the same source of infection such as a contaminated water well. The epidemic curve will be broad.

2-Propagated epidemic: an epidemic in which the infectious agent continues to spread from persons to person. In this case the infection will last for a longer time and there will be several peaks in the number of the cases. Measles and chickenpox can start as a common source epidemic in a school and continue as a propagated epidemic.

The Epidemic Curve





The Epidemic Curve

Time of infection

Number
of
cases

time

Minimum
incubation
period

Maximum incubation period



Conclusions

1-Components of communicable diseases are:

Agent, Host and Environment

2-Infectious diseases in order to success & establish:

The agent should be multiplying, have a susceptible host ,exist from that host and infect another host

3-Epidemiologic classification that based on the methods of transmission of the diseases are:

Contact , Inhalation , Ingested food & water,
Penetration of the skin by vector and transplacental



- 4-Transmission of the agent to a new host depend on host factors
- 5- **Carrier can be defined as human** get M.O. but with no signs & symptoms
- 7-Time periods of infection divided into time of infection, latent period, infectious period or communicable period and non-infectious period,while Time periods of isease divided into time of infection, the incubation period, symptomatic period and a symptomatic period



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9-Types of epidemics : Common source epidemic (Point source and Extended source outbreak) and Propagated epidemic



Epidemiology of Communicable Diseases – MCQs with Answers & Explanations

1 ☐ Basic Concepts

1. Which of the following best defines an infectious disease?

- A) A disease spread only by insects
- B) Any disease caused by an infective agent
- C) Any genetic disease
- D) Only viral diseases



Answer: B

Explanation: An infectious disease is caused by an infective agent (bacteria, virus, fungus, etc.), regardless of how it spreads.

2. Which of the following best defines a communicable disease?

- A) A noninfectious illness
- B) Any infectious disease that spreads from one person to another
- C) A disease caused by toxins only
- D) A disease that affects animals only



Answer: B

Explanation: A communicable disease is an infectious disease that can be transmitted between individuals.

3. What are the three components of a communicable disease known as the epidemiologic triad?

- A) Agent, Host, Environment
- B) Pathogen, Vector, Host
- C) Host, Immunity, Climate
- D) Agent, Vector, Host

 Answer: A

Explanation: The classic triad: Agent (cause), Host (susceptible person), and Environment (conditions allowing spread).

4. The distribution of a communicable disease is described using:

- A) Person, Place, and Time
- B) Cause, Effect, and Treatment
- C) Host, Agent, and Environment
- D) Diagnosis, Therapy, and Outcome

 Answer: A

Explanation: Epidemiologists describe disease patterns by who (person), where (place), and when (time).

5. Infection can exist in which two main forms?

- A) Clinical and Subclinical
- B) Acute and Chronic
- C) Mild and Severe
- D) Endemic and Pandemic

✓ Answer: A

Explanation: Subclinical means no visible symptoms; clinical means symptomatic infection.

6. Which of the following is NOT essential for the establishment of an infectious disease?

- A) Agent multiplication
- B) Presence of susceptible host
- C) Agent's exit from host
- D) Agent cannot infect another host

✓ Answer: D

Explanation: For an infection to spread, the agent must infect another host – so inability to infect another host prevents spread.

2 ☐ Characteristics of Communicable Diseases

7. Communicable diseases are caused by:

- A) Nutritional deficiencies
- B) Infectious agents or their toxic products
- C) Genetic mutations
- D) Environmental toxins

✓ Answer: B

Explanation: The cause is a microorganism or its toxic products (virus, bacteria, protozoa, helminth, fungus).

8. Malnutrition makes a person more susceptible to:

- A) Measles
- B) Tetanus
- C) Hypertension
- D) Cancer

 Answer: A

Explanation: Malnutrition weakens immunity, making infections like measles more severe.

9. Fatigue affects infection susceptibility by:

- A) Strengthening immunity
- B) Weakening immunity
- C) Preventing exposure
- D) Killing pathogens

 Answer: B

Explanation: Fatigue depresses immune function and increases susceptibility.

10. Pregnant women are more liable to infection because of:

- A) Reduced body temperature
- B) Hormonal and immune changes
- C) Dehydration
- D) Malnutrition only

☒ Answer: B

Explanation: Pregnancy alters immune response and physiology, increasing risk of infection.

3 ☐ Reservoirs & Carriers

11. A reservoir is defined as:

- A) A person who has recovered from infection
- B) A place where an agent multiplies and survives
- C) A diagnostic laboratory
- D) A contaminated object

☒ Answer: B

Explanation: A reservoir supports the survival and multiplication of the pathogen.

12. Which of the following is NOT a type of reservoir?

- A) Human
- B) Animal
- C) Non-living
- D) Genetic

☒ Answer: D

Explanation: Genetic material is not a reservoir; the other three are true types.

13. A person who harbors microorganisms but shows no symptoms and transmits disease is called a:

- A) Host
- B) Carrier
- C) Case
- D) Vector

✅ Answer: B

Explanation: Carriers are asymptomatic individuals capable of spreading disease.

14. Carriers are epidemiologically important because:

- A) They cannot transmit disease
- B) They are immune to infection
- C) They silently spread infection
- D) They are rare

✅ Answer: C

Explanation: Carriers have no symptoms but can infect others – making them difficult to detect and control.

15. Which disease often has chronic carriers?

- A) Measles
- B) Typhoid (Salmonella)
- C) Influenza
- D) COVID-19

✅ Answer: B

Explanation: Typhoid fever may have chronic carriers who shed bacteria intermittently.

16. A zoonotic disease is one that:

- A) Affects only humans
- B) Transmits from vertebrate animals to humans
- C) Is caused by parasites only
- D) Occurs only in tropical areas

☒ Answer: B

Explanation: Zoonoses are diseases transmitted from animals to humans (e.g., rabies).

17. Example of a zoonotic disease:

- A) Cholera
- B) Rabies
- C) Measles
- D) Tuberculosis

☒ Answer: B

Explanation: Rabies spreads from infected animals (dogs, bats) to humans.

4 ☐ Host & Environmental Factors

18. Which factor is NOT related to host immunity?

- A) Age
- B) Sex
- C) Weather
- D) Nutrition

✓ Answer: C

Explanation: Weather is an environmental, not host, factor.

19. Poor sanitation and poverty mainly affect:

- A) Agent virulence
- B) Transmission of disease
- C) Genetic mutation
- D) Incubation period

✓ Answer: B

Explanation: Unclean environments and close contact increase transmission rates.

20. Education level influences disease occurrence by:

- A) Directly changing immunity
- B) Affecting hygiene and prevention behavior
- C) Altering climate
- D) Increasing virulence

✓ Answer: B

Explanation: Education improves awareness, hygiene, and preventive practices.

5 ☐ Modes of Transmission

21. Which is an example of direct contact transmission?

- A) Using contaminated tools
- B) Shaking hands with an infected person
- C) Drinking contaminated water
- D) Inhaling dust particles

☒ Answer: B

Explanation: Direct contact includes physical contact (touch, sexual contact, etc.).

22. Which is an example of indirect contact transmission?

- A) Hugging
- B) Sharing a pen or cup
- C) Breathing near an infected person
- D) Mosquito bite

☒ Answer: B

Explanation: Indirect transmission involves inanimate objects (fomites).

23. Inhalation transmission includes:

- A) Droplet and airborne routes
- B) Fomite and foodborne routes
- C) Vector and transplacental routes
- D) Skin and wound routes

✓ Answer: A

Explanation: Droplets (direct) and airborne particles (indirect) transmit respiratory infections.

24. Food and water transmission are examples of:

- A) Contact
- B) Inhalation
- C) Ingestion
- D) Penetration

✓ Answer: C

Explanation: Ingested pathogens cause fecal-oral diseases like cholera or typhoid.

25. Which is an example of vector transmission?

- A) Sharing needles
- B) Mosquito-borne malaria
- C) Sneezing
- D) Breastfeeding

✓ Answer: B

Explanation: Vectors (e.g., mosquitoes, fleas) carry pathogens between hosts.

26. Transplacental transmission refers to:

- A) Spread through contaminated needles
- B) Spread from mother to fetus
- C) Airborne spread
- D) Spread via food

✅ Answer: B

Explanation: Certain infections (e.g., rubella, HIV) can cross the placenta.

Part 2 – Time Periods of Infection & Disease

1. The time of infection refers to:

- A) When symptoms first appear
- B) When the infectious agent first enters the susceptible individual
- C) When the immune response starts
- D) When recovery begins

✅ Answer: B

Explanation: The time of infection is the exact moment the microorganism enters the host's body.

2. The latent period is the time between:

- A) Infection and symptom appearance

- B) Infection and ability to transmit infection
- C) Symptom onset and recovery
- D) Infection and death

✅ Answer: B

Explanation: The latent period is when the agent is present but not yet transmissible. After this, the person becomes infectious.

3. In measles, the latent period is:

- A) Equal to the incubation period
- B) Longer than the incubation period
- C) Shorter than the incubation period
- D) Absent

✅ Answer: C

Explanation: In measles, the person becomes infectious before showing symptoms, meaning the latent period is shorter.

4. The infectious (communicable) period is defined as:

- A) The period when infection is first acquired
- B) The time when the infected person can transmit the disease
- C) The time after recovery
- D) The time before symptoms

✅ Answer: B

Explanation: The infectious period is when a person can spread the pathogen to others.

5. For diseases like measles and hepatitis, the infectious period begins:

- A) After symptoms appear
- B) Before symptoms appear
- C) Only after recovery
- D) At death

 Answer: B

Explanation: Measles and hepatitis patients become infectious even before they show symptoms.

6. For cholera, the infectious period starts:

- A) Before symptoms
- B) After symptoms begin
- C) During incubation
- D) During convalescence

 Answer: B

Explanation: In cholera, bacteria are shed after the person starts showing symptoms (profuse diarrhea).

7. The non-infectious period occurs when:

- A) The person recovers or dies
- B) The infection starts
- C) Symptoms first appear
- D) Transmission peaks

✓ Answer: A

Explanation: The non-infectious period begins when the infected person is no longer capable of transmitting the disease (due to recovery, immunity, or death).

8. Which statement about infection vs. disease is TRUE?

- A) Infection always leads to disease
- B) All infections are symptomatic
- C) Infection can exist without symptoms
- D) Infection and disease mean the same thing

✓ Answer: C

Explanation: Not all infections cause symptoms – some are asymptomatic infections.

9. The incubation period is:

- A) Time from infection to symptom development
- B) Time from infection to infectiousness
- C) Time from symptoms to recovery
- D) Time from infection to death

✓ Answer: A

Explanation: The incubation period is the duration between infection and onset of symptoms.

10. Factors affecting the incubation period include all EXCEPT:

- A) Nature of the pathogen
- B) Infective dose
- C) Portal of entry
- D) Educational level of the host

✓ Answer: D

Explanation: Incubation period depends on biological factors, not social ones.

11. The symptomatic period refers to:

- A) The period of recovery
- B) The period during which the patient shows signs and symptoms
- C) The incubation period
- D) The latent period

✓ Answer: B

Explanation: The symptomatic period is when the disease manifests clinically.

12. The asymptomatic period refers to:

- A) The time after recovery when no symptoms are present
- B) The time before infection
- C) The incubation period
- D) The same as the latent period

✓ Answer: A

Explanation: After symptoms end, the asymptomatic period begins –

the person may recover, die, or become immune.

13. A person who recovers but continues to spread infection is called a:

- A) Carrier
- B) Reservoir
- C) Host
- D) Vector

✓ Answer: A

Explanation: Carriers can continue transmitting the infection even after recovery (e.g., typhoid carriers).

14. Which period describes when an individual can transmit infection without showing symptoms?

- A) Incubation period
- B) Latent period
- C) Infectious period
- D) Both A and C

✓ Answer: D

Explanation: During incubation (for diseases like measles), the person is infectious even before symptoms appear.

15. Why is the relationship between latent and incubation periods important in public health?

- A) It helps calculate infection fatality rates
- B) It determines when a person becomes contagious relative to symptoms
- C) It helps estimate recovery time
- D) It determines which antibiotics to use

✓ Answer: B

Explanation: Knowing whether a disease spreads before symptoms helps in designing isolation and control strategies.

16. In measles, isolating children only after symptoms appear fails to stop all transmission because:

- A) The virus mutates quickly
- B) They are infectious before symptoms appear
- C) The disease has no treatment
- D) Vaccines don't work

✓ Answer: B

Explanation: Measles patients are contagious during the incubation period – before the rash or fever begins.

17. The epidemic curve helps in identifying:

- A) Incubation period range
- B) Infectious period
- C) Time of exposure
- D) All of the above

✓ Answer: D

Explanation: The shape of the epidemic curve provides clues about exposure timing, incubation range, and transmission type.

Part 3 – Types of Epidemics

1. An epidemic is best defined as:

- A) The constant presence of a disease within a region
- B) A sudden increase in disease occurrence clearly above normal expectations
- C) A disease that spreads globally
- D) An infection that affects only animals

 Answer: B

Explanation: An epidemic occurs when a disease appears in numbers clearly higher than expected in a given area or population.

2. When a disease remains constantly present in a population at a steady rate, it is called:

- A) Epidemic
- B) Endemic
- C) Pandemic
- D) Sporadic

 Answer: B

Explanation: Endemic means the usual, constant presence of a disease within a certain region (e.g., malaria in tropical areas).

3. When a rare disease appears with even a few cases, it may be considered:

- A) An endemic
- B) An epidemic
- C) A pandemic
- D) A sporadic case

✓ Answer: B

Explanation: For rare diseases, even a few cases can be an epidemic because the expected number is near zero.

4. A common source epidemic occurs when:

- A) People are infected one after another
- B) All infections come from the same source
- C) Disease is genetically transmitted
- D) The disease spreads globally

✓ Answer: B

Explanation: In a common source epidemic, everyone is exposed to the same origin – like contaminated water or food.

5. Example of a common source epidemic:

- A) Influenza outbreak

- B) Food poisoning at a wedding
- C) Measles in a school
- D) Chickenpox in a hospital

✅ Answer: B

Explanation: A single contaminated meal infecting multiple people is a point source epidemic (a type of common source).

6. When everyone is exposed to the same source only once, the epidemic is called a:

- A) Propagated epidemic
- B) Extended source epidemic
- C) Point source epidemic
- D) Pandemic

✅ Answer: C

Explanation: Point source epidemics result from a single exposure event (e.g., one contaminated party meal).

7. In a point source epidemic, the epidemic curve typically looks:

- A) Flat and wide
- B) Sharp and bell-shaped
- C) Repetitive with multiple peaks
- D) Constant with no change

✅ Answer: B

Explanation: All cases occur close together in time, producing a sharp bell-shaped curve.

8. When exposure to the source is repeated over time, it is called:

- A) Point source epidemic
- B) Extended source outbreak
- C) Propagated epidemic
- D) Pandemic

✅ Answer: B

Explanation: Extended source epidemics occur with continuous exposure, such as a contaminated well used daily.

9. The epidemic curve for an extended source outbreak is:

- A) Narrow and sharp
- B) Flat and broad
- C) Multiple peaks
- D) Declining line

✅ Answer: B

Explanation: Because exposure continues, new cases occur over an extended period → a broad, flatter curve.

10. A propagated epidemic occurs when:

- A) Infection spreads person to person
- B) Everyone gets infected from one meal
- C) Disease is confined to one place
- D) It spreads through food only

✓ Answer: A

Explanation: In propagated epidemics, transmission is from one infected person to another – e.g., measles or chickenpox.

11. In a propagated epidemic, the curve shows:

- A) One single peak
- B) Several peaks over time
- C) No increase
- D) A random pattern

✓ Answer: B

Explanation: Because infection spreads in waves (each wave infecting new people), there are multiple peaks on the epidemic curve.

12. Which of the following diseases can start as a common source epidemic and continue as a propagated one?

- A) Typhoid
- B) Cholera
- C) Measles
- D) Rabies

✓ Answer: C

Explanation: Measles and chickenpox can begin with a shared exposure and then spread person to person, making it propagated.

13. A pandemic is best described as:

- A) A disease outbreak confined to one family
- B) An epidemic that spreads to multiple countries or continents
- C) A constant level of infection in one area
- D) A short, localized outbreak

 Answer: B

Explanation: A pandemic is a global epidemic (e.g., COVID-19, 1918 influenza).

14. Which of the following is an example of a pandemic?

- A) Cholera outbreak in one village
- B) Seasonal flu in one country
- C) COVID-19 outbreak across multiple continents
- D) Chickenpox in a school

 Answer: C

Explanation: COVID-19 spread worldwide, fulfilling the definition of a pandemic.

15. What key feature differentiates propagated epidemics from common source epidemics?

- A) Number of cases
- B) Duration of exposure
- C) Mode of transmission
- D) Incubation period

 Answer: C

Explanation: Common source → exposure to one source; Propagated → transmission between people.

16. Which type of epidemic usually lasts the longest?

- A) Point source
- B) Extended source
- C) Propagated epidemic
- D) Endemic

✓ Answer: C

Explanation: Person-to-person spread continues over time, making propagated epidemics the longest lasting.

17. The epidemic curve helps determine:

- A) Type of epidemic
- B) Period of exposure
- C) Incubation period range
- D) All of the above

✓ Answer: D

Explanation: The curve's shape shows how an epidemic spreads, when exposure occurred, and how long incubation lasts.

18. The first case in an epidemic is known as:

- A) Secondary case

- B) Carrier case
- C) Index case
- D) Sentinel case

✅ Answer: C

Explanation: The index case is the first recognized case in an outbreak and helps trace the source of infection.

19. If a disease occurs predictably every year (like flu season), it is best described as:

- A) Sporadic
- B) Endemic with periodic increase
- C) Propagated
- D) Pandemic

✅ Answer: B

Explanation: Seasonal patterns represent endemicity with periodic peaks.

20. When an epidemic spreads globally, affecting millions, it is called a:

- A) Endemic
- B) Propagated epidemic
- C) Pandemic
- D) Common source outbreak

✅ Answer: C

Explanation: A pandemic is simply an epidemic on a global scale.

✅ That covers the entire PDF – all core sections:

1. Basics of communicable disease
2. Host-agent-environment
3. Reservoirs & carriers
4. Modes of transmission
5. Time periods (infection & disease)
6. Types of epidemics

🧩 Epidemiology of Communicable Diseases – Full Practice MCQ Test (30 Questions)

1. Which of the following best defines a communicable disease?

- A) A disease caused by genetics
- B) Any infectious disease transmitted from one person to another
- C) A disease spread only by vectors
- D) A noninfectious illness

2. The three main components of a communicable disease are:

- A) Host, Agent, Environment
- B) Host, Immunity, Transmission
- C) Agent, Vector, Host

D) Agent, Treatment, Environment

3. "Person, Place, and Time" are used to describe:

- A) Disease distribution
 - B) Clinical symptoms
 - C) Genetic risk
 - D) Laboratory diagnosis
-

4. A person who harbors infection without symptoms but can transmit it is called a:

- A) Host
 - B) Carrier
 - C) Reservoir
 - D) Vector
-

5. A zoonotic disease refers to:

- A) Human-to-human transmission
 - B) Transmission from animals to humans
 - C) Hospital-acquired infection
 - D) Genetically inherited disease
-

6. Which factor does NOT affect host immunity?

- A) Age
- B) Nutrition

C) Climate

D) Sex

7. Malnutrition increases susceptibility to:

A) Diabetes

B) Measles

C) Tuberculosis

D) Rabies

8. The place where an organism can survive and multiply is called:

A) Host

B) Reservoir

C) Vector

D) Environment

9. A chronic carrier is best illustrated by which disease?

A) Typhoid

B) Influenza

C) Measles

D) COVID-19

10. Direct contact transmission includes:

A) Shaking hands

B) Sharing a pen

- C) Drinking contaminated water
 - D) Mosquito bites
-

11. Indirect contact occurs via:

- A) Airborne droplets
 - B) Fomites
 - C) Kissing
 - D) Transplacental spread
-

12. Transmission from mother to fetus is called:

- A) Vector-borne
 - B) Airborne
 - C) Transplacental
 - D) Indirect
-

13. The latent period is the time between:

- A) Infection and appearance of symptoms
 - B) Infection and becoming infectious
 - C) Symptoms and recovery
 - D) Infection and death
-

14. The incubation period is defined as:

- A) Infection to symptom onset
- B) Infection to infectiousness

- C) Recovery to death
 - D) Symptom to recovery
-

15. The infectious (communicable) period refers to:

- A) Time when disease can be transmitted
 - B) Time before infection
 - C) Time after death
 - D) Time before immunity develops
-

16. In measles, the patient becomes infectious:

- A) Only after symptoms
 - B) Before symptoms appear
 - C) Only during convalescence
 - D) After recovery
-

17. Which period describes when symptoms are present?

- A) Latent
 - B) Incubation
 - C) Symptomatic
 - D) Asymptomatic
-

18. When a person recovers but continues to spread disease, he is called a:

- A) Host

- B) Carrier
 - C) Reservoir
 - D) Vector
-

19. When a disease rate is constant in a region, it is called:

- A) Epidemic
 - B) Endemic
 - C) Pandemic
 - D) Sporadic
-

20. An epidemic means:

- A) Constant occurrence of a disease
 - B) A sudden increase above expected levels
 - C) A disease limited to animals
 - D) A disease confined to hospitals
-

21. When everyone is infected from a single contaminated meal, it is a:

- A) Point source epidemic
 - B) Extended source outbreak
 - C) Propagated epidemic
 - D) Endemic
-

22. When exposure to the source is continuous, the epidemic is:

- A) Point source

- B) Extended source
 - C) Propagated
 - D) Pandemic
-

23. A propagated epidemic spreads:

- A) Through person-to-person transmission
 - B) From one contaminated source
 - C) Through water only
 - D) Through soil only
-

24. In a propagated epidemic, the epidemic curve typically shows:

- A) A single sharp peak
 - B) Several waves of peaks
 - C) A flat line
 - D) Random distribution
-

25. A pandemic is:

- A) A global epidemic
 - B) A local outbreak
 - C) A hospital infection
 - D) A seasonal disease
-

26. Which of the following best describes an extended source epidemic curve?

- A) Sharp and short
 - B) Broad and prolonged
 - C) Flat with no cases
 - D) Random spikes
-

27. The index case is:

- A) The first identified case in an epidemic
 - B) The most severe case
 - C) The last case to recover
 - D) A secondary infection
-

28. Poor sanitation and crowding mainly affect:

- A) Immunity
 - B) Transmission
 - C) Genetic risk
 - D) Incubation period
-

29. A person who gets infection but shows no symptoms is called:

- A) A clinical case
 - B) A subclinical case
 - C) A secondary case
 - D) A reservoir
-

30. The relation between latent and incubation periods is important

because it:

- A) Helps estimate recovery time
 - B) Shows when transmission occurs relative to symptoms
 - C) Predicts mortality
 - D) Determines the agent's type
-

Answer Key & Quick Explanations

- 1 ☐ B - Communicable = transmissible infection
- 2 ☐ A - Agent, Host, Environment
- 3 ☐ A - Describes disease patterns
- 4 ☐ B - Carriers transmit without symptoms
- 5 ☐ B - Zoonotic = animal → human
- 6 ☐ C - Climate = environmental factor
- 7 ☐ B - Malnutrition worsens measles
- 8 ☐ B - Reservoir = place agent survives
- 9 ☐ A - Typhoid can have chronic carriers
- 10 ☒ A - Direct = physical contact
- 1 1 ☐ B - Indirect = via fomites (objects)
- 1 2 ☐ C - Transplacental = mother → fetus
- 1 3 ☐ B - Latent = before becoming infectious
- 1 4 ☐ A - Incubation = infection → symptoms
- 1 5 ☐ A - Infectious = capable of transmission
- 1 6 ☐ B - Measles infectious before symptoms
- 1 7 ☐ C - Symptomatic = showing signs
- 1 8 ☐ B - Carriers spread post-recovery
- 1 9 ☐ B - Endemic = stable, constant rate

- 20 ☐ B - Epidemic = above normal expectation
- 21 ☐ A - Point source = single exposure
- 22 ☐ B - Extended = continuous exposure
- 23 ☐ A - Propagated = person-to-person
- 24 ☐ B - Several peaks (waves of infection)
- 25 ☐ A - Pandemic = global spread
- 26 ☐ B - Broad curve = ongoing exposure
- 27 ☐ A - Index case = first detected case
- 28 ☐ B - Transmission ↑ with poor sanitation
- 29 ☐ B - Subclinical = infected but no signs
- 30 ☐ B - Shows if people are infectious before symptoms
